EDUSAMBAM ASSESSMENT SERIES

Biology Comprehensive Practice Test 4

PRINTABLE PRACTICE WORKSHEET

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Candidate Name:		**Roll No.:**	
Class / Grade:		**Date:**	
School / Institution:		**Test Centre:**	

Subject	Biology	**Topic**	Biology Comprehensive Practice Test 4	**Total Marks**	100
Total Questions	45	**Time Allowed**	1 Hour (60 minutes)	**Version**	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	**45**	**—**	**100**

Section A — Multiple Choice — 30 marks

1. Bacteria are generally classified as: — Basic · 2 pts

A. Eukaryotic animals
B. Viruses with nuclei
C. Prokaryotic cells
D. Multicellular fungi

Answer: ☐A ☐B ☐C ☐D

2. Viruses are best described as: — Basic · 2 pts

A. Independent cells with ribosomes that divide on their own
B. Acellular infectious agents that depend on host cells to replicate
C. Photosynthetic bacteria found in water
D. Multicellular parasites that live in the gut

Answer: ☐A ☐B ☐C ☐D

3. A pathogen is: — Basic · 2 pts

A. Any harmless microorganism found on the skin
B. A white blood cell that fights infection
C. A vaccine ingredient in every case
D. An organism or agent capable of causing disease

Answer: ☐A ☐B ☐C ☐D

4. Which statement about malaria is correct? — Basic · 2 pts

A. It is caused by a single-celled protist (Plasmodium) carried by Anopheles mosquitoes
B. It is caused by a bacterium spread through contaminated water
C. It is caused by a virus spread by respiratory droplets
D. It is caused by a fungus spread by direct skin contact

Answer: ☐A ☐B ☐C ☐D

5. Which of these diseases is caused by a virus? **Basic · 2 pts**

A. Tuberculosis
B. Malaria
C. Influenza
D. Athlete's foot

Answer: ☐A ☐B ☐C ☐D

6. Which measure most directly reduces the spread of cholera? **Intermediate · 2 pts**

A. Providing clean drinking water and safe disposal of sewage
B. Taking antibiotics every day as a precaution
C. Avoiding all contact with other people for life
D. Eating only cooked starchy foods

Answer: ☐A ☐B ☐C ☐D

7. Phagocytes help defend the body by: **Intermediate · 2 pts**

A. Producing red blood cells in the bone marrow
B. Making the hormone insulin
C. Carrying oxygen using haemoglobin
D. Engulfing and digesting microbes or debris

Answer: ☐A ☐B ☐C ☐D

8. The incubation period is the time between: **Intermediate · 2 pts**

A. Vaccination and the production of antibodies
B. Infection and the appearance of the first symptoms
C. The first symptom and full recovery
D. Recovery and the next infection

Answer: ☐A ☐B ☐C ☐D

9. A chromosome is best described as: **Intermediate · 2 pts**

A. DNA associated with proteins and organised into a structural unit
B. A type of carbohydrate in the cell wall
C. A membrane-bound organelle without DNA
D. A single protein molecule

Answer: ☐A ☐B ☐C ☐D

10. A gene is best understood as: **Intermediate · 2 pts**

A. An entire organism
B. A cell organelle
C. A functional DNA sequence that contributes to a biological product or its regulation
D. A droplet of stored lipid

Answer: ☐A ☐B ☐C ☐D

11. DNA replication is described as semiconservative because: **Advanced · 2 pts**

A. Both new DNA strands are made entirely of new material
B. Each daughter DNA molecule has one parental strand and one new strand
C. No parental DNA is kept in either daughter molecule
D. Only the proteins of the chromosome are copied

Answer: ☐A ☐B ☐C ☐D

12. Meiosis is important in sexual reproduction because it: **Advanced · 2 pts**

A. Produces identical diploid body cells only
B. Stops chromosomes from separating
C. Removes all genetic variation
D. Produces haploid cells and contributes to genetic variation

Answer: ☐A ☐B ☐C ☐D

13. A mutation is: **Advanced · 2 pts**

A. A change in the sequence or structure of genetic material
B. Always harmful to the organism
C. Always beneficial to the organism
D. Always caused by exercise or diet

Answer: ☐A ☐B ☐C ☐D

14. Both parents are heterozygous carriers (Aa) of a recessive allele. What is the chance that a child inherits two recessive alleles (aa)? **Advanced · 2 pts**

A. 0%, because both parents are unaffected
B. 50%
C. 25% (1 in 4)
D. 100%

Answer: ☐A ☐B ☐C ☐D

15. A phenotype is influenced by: **Advanced · 2 pts**

A. Genes only, in every case
B. Genotype and environment, depending on the trait
C. Environment only, in every case
D. Chromosome number only

Answer: ☐A ☐B ☐C ☐D

Section B — True / False **20 marks**

16. All microorganisms cause disease. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

17. Some bacteria are beneficial to humans and ecosystems. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

18. Viruses generally require host cells to reproduce. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

19. A pathogen must always cause symptoms immediately after entering a host. Basic · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

20. Washing hands with soap can reduce the spread of some infections. Basic · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

21. DNA contains information encoded in the sequence of its nucleotides. Intermediate · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

22. Every mutation causes a visible disease. Intermediate · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

23. Meiosis normally produces genetically identical daughter cells. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

24. A dominant allele is always more common in a population than a recessive allele. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

25. Genetic information can influence traits without being the only influence on every trait. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

Section C — Fill in the Blank 20 marks

26. Viruses are generally described as _____ because they lack cellular organisation. Intermediate · 2 pts
Answer: _____

27. The spread of a pathogen from one host to another is called _____. Intermediate · 2 pts
Answer: _____

28. An animal such as a mosquito that carries a pathogen from one host to another is called a _____. Intermediate · 2 pts
Answer: _____

29. Bacteria are _____ organisms because their cells have no nucleus. Intermediate · 2 pts
Answer: _____

30. The time between infection and the first symptoms is called the _____ period. Intermediate · 2 pts
Answer: _____

31. A cell containing two sets of chromosomes is described as _____. Advanced · 2 pts
Answer: _____

32. The process by which the information in a gene is used to make a functional product is gene _____. Advanced · 2 pts
Answer: _____

33. The genetic make-up of an organism is its _____. Advanced · 2 pts
Answer: _____

34. The observable characteristics of an organism form its _____. Advanced · 2 pts
Answer: _____

35. Exchange of DNA between homologous chromosomes during meiosis is called _____ over. Advanced · 2 pts
Answer: _____

Section D — Column Matching — 10 marks

36–40. Match each term in Column A with the correct description in Column B. **Advanced · 10 pts**

Column A	Answer	Column B
Bacterium	____	**A.** Functional region of DNA
Virus	____	**B.** Carries a pathogen between hosts
Vector	____	**C.** DNA packaged with proteins
Gene	____	**D.** Prokaryotic microorganism
Chromosome	____	**E.** Acellular infectious agent

Section E — Short Answer — 20 marks

41. Explain why the presence of a microorganism does not automatically mean that disease will occur. **Advanced · 4 pts**

42. Why can respiratory infections spread efficiently even when an infected person does not have severe symptoms? **Advanced · 4 pts**

43. Explain the relationship among DNA, genes and chromosomes. **Advanced · 4 pts**

44. Explain why meiosis is important for both chromosome-number stability and genetic variation in sexual reproduction. **Advanced · 4 pts**

45. A child has a recessive condition while both parents are unaffected. Explain a plausible inheritance pattern. **Advanced · 4 pts**

Candidate Signature: _____ **Invigilator Signature:** _____

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Marks Obtained: / 100	Checked By:
Result / Grade:	Date Checked:

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Biology Comprehensive Practice Test 4 — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

1. C
2. B
3. D
4. A
5. C
6. A
7. D
8. B
9. A
10. C
11. B
12. D
13. A
14. C
15. B

Section B — True / False

16. False
17. True
18. True
19. False
20. True
21. True
22. False
23. False
24. False
25. True

Section C — Fill in the Blank (accepted answers)

26. acellular
27. transmission
28. vector / vectors
29. prokaryotic / prokaryote / prokaryotes
30. incubation
31. diploid
32. expression
33. genotype
34. phenotype
35. crossing

Section D — Column Matching

36. Bacterium — D
37. Virus — E
38. Vector — B
39. Gene — A
40. Chromosome — C

Section E — Short Answer (Model Answers)

41. Many microorganisms are harmless or helpful, such as normal gut bacteria. Even a pathogen causes disease only if enough of it enters by a suitable route, multiplies and causes damage, and if the host's defences, such as skin barriers and the immune system, do not stop it first.
Marking: 1 mark: many microorganisms are harmless or helpful; 1 mark: the number (dose) may be too small; 1 mark: the host's defences (barriers, immune system) may stop it; 1 mark: a pathogen must enter, multiply and cause damage.

42. People can be infectious before symptoms start or with only mild symptoms, so they often do not know they are infected. They keep meeting others, and the pathogen travels in respiratory droplets or aerosols made when they breathe, talk or cough, especially in close, crowded or poorly ventilated places.
Marking: 1 mark: people may be infectious before symptoms or with only mild symptoms; 1 mark: pathogens travel in droplets or aerosols from breathing, talking or coughing; 1 mark: people are unaware and keep mixing with others; 1 mark: close, crowded or poorly ventilated places help spread.

43. DNA is the molecule that stores hereditary information in the sequence of its nucleotides. A gene is a section of DNA that carries the instructions for making a particular product, usually a protein. DNA is wound with proteins and packaged into chromosomes, and each chromosome contains many genes.
Marking: 1 mark: DNA is the molecule that stores hereditary information in its sequence; 1 mark: a gene is a section of DNA that codes for a product; 1 mark: DNA is packaged with proteins into chromosomes; 1 mark: each chromosome contains many genes.

44. Meiosis halves the chromosome number to make haploid gametes, and fertilisation then restores the diploid number, so the number stays constant from generation to generation. Crossing over and the random assortment of chromosomes during meiosis create new combinations of alleles, so offspring differ genetically.
Marking: 1 mark: meiosis halves the chromosome number (haploid gametes); 1 mark: fertilisation restores the diploid number, so it stays constant; 1 mark: crossing over exchanges DNA between homologous chromosomes; 1 mark: random assortment of chromosomes gives new combinations.

45. Both parents are probably carriers: each is heterozygous with one dominant and one recessive allele, so they are unaffected because the dominant allele masks the recessive one. Each parent passed on a recessive allele, so the child inherited two copies (homozygous recessive). Each child of two carriers has a 1 in 4 (25%) chance of this.
Marking: 1 mark: both parents are carriers (heterozygous); 1 mark: each parent passes on a recessive allele; 1 mark: the child inherits two recessive copies (homozygous recessive); 1 mark: the dominant allele masks the recessive one in the parents; each child has a 1 in 4 chance.